

Midterm Solutions

ECON 301
Spring 2022



Problem 1

(a) Let $Y_i = \text{finalscore}_i$, and $X_i = \text{percentattend}_i$. The model according to A2 is $E[Y_i | X_i] = \beta_0 + \beta_1 X_i$. This β_1 is the ceteris paribus effect of the percentage of classes attended on final score. This effect is estimated as $\hat{\beta}_1 = 0.09672$, that is, an increase of 1% in attendance increases the final score by 0.09672 points.

(b) $H_0: \beta_1 = 0$ vs $H_1: \beta_1 \neq 0$. The test statistic is $T = \frac{\hat{\beta}_1}{SE(\hat{\beta}_1)} = \frac{0.09672}{0.02627} \approx 3.68177$ and $T \sim t(678)$

Using the table in the back, the critical value for a two sided test at the 5% level is approx. (using the last row) $t_{0.025}^{678} \approx 1.96$. Since $|T| = 3.68177 > 1.96$ we reject the null hypothesis. Thus the effect is statistically significant. However, as the answer in (a) says, it is too small (0.09672 points) to be of practical relevance.

(c) The effect ΔY is given by the expression $\Delta Y = \beta_1 \Delta X \approx 20\beta_1$, which is estimated via $\hat{\Delta Y} = 20\hat{\beta}_1 = 1.9344$. Thus, the hypothesis to test is: $H_0: \Delta Y = 2$ vs $H_1: \Delta Y < 2$, which is a left-sided test. We will use a T statistic for the test; to do this we need $SE(20\hat{\beta}_1) = \sqrt{20^2 \text{Var}(\hat{\beta}_1)} = 20 SE(\hat{\beta}_1) = 0.5254$. Thus: $T = \frac{1.9344 - 2}{0.5254} = -0.124857$

Again, we have that $T \sim t(678)$ so the critical value for a left-sided test at the 1% level is approx. (using the last row of the table)

$t_{0.01}^{678} \approx -2.326$. Since $-2.326 < -0.124857$ we cannot reject H_0 at the 1%.

(d) Recall that $\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$, so that $\bar{Y} = \hat{\beta}_0 + \hat{\beta}_1 \bar{X}$ and thus the predicted score at $\bar{X}$ is $\bar{Y} = 64.72794$

(e) That is $\hat{\sigma}^2 = \hat{\text{Var}}(Y|X) = 11.67^2 = 136.189$

(f) $\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 \cdot 100 = 66.4968$

$$(g) \cdot E(\hat{\beta}_0 + 100\hat{\beta}_1 | X's) = \beta_0 + 100\beta_1.$$

$$\cdot \text{Var}(\hat{\beta}_0 + 100\hat{\beta}_1 | X's) = \text{Var}(\hat{\beta}_0 | X's) + 10,000 \cdot \text{Var}(\hat{\beta}_1 | X's) + 200 \text{Cov}(\hat{\beta}_0, \hat{\beta}_1 | X's)$$

These are the formulas. To estimate them we use our estimators:

$$\cdot \hat{\beta}_0 + 100\hat{\beta}_1 = 66.4968$$

$$\begin{aligned} & \cdot \hat{\text{Var}}(\hat{\beta}_0 | X's) + 10,000 \cdot \hat{\text{Var}}(\hat{\beta}_1 | X's) + 200 \text{Cov}(\hat{\beta}_0, \hat{\beta}_1 | X's) \\ & = 4.8060 + 10,000(0.00069) + 200(-0.05637) \\ & = 0.432. \end{aligned}$$

$$(h) \text{ Note that } SE(\hat{Y}) = SE(\hat{\beta}_0 + 100\hat{\beta}_1) = \sqrt{\hat{\text{Var}}(\hat{\beta}_0 + 100\hat{\beta}_1)} \\ = \sqrt{0.432} = 0.657267$$

Thus we can use the T statistic as our pivot:

$$T = \frac{\hat{Y} - (\beta_0 + 100\beta_1)}{SE(\hat{Y})} = \frac{66.4968 - (\beta_0 + 100\beta_1)}{0.657267}$$

Once again $T \sim t(678)$, and therefore the CI can be constructed as follows:

$$\Pr(|T| \leq t_{0.025}^{678}) = 0.95 \Leftrightarrow$$

$$\Pr(\hat{Y} - t_{0.025}^{678} \cdot SE(\hat{Y}) \leq \beta_0 + 100\beta_1 \leq \hat{Y} + t_{0.025}^{678} \cdot SE(\hat{Y})) = 0.95$$

The true effect on final score at 100% attendance

Using the table $t_{0.025}^{678} \approx 1.960$ and thus the

CI bounds are given by $66.4968 \pm 1.96 (0.657267)$,

finishing the calculation we have:

$$CI = [65.2086, 67.785].$$

(i) Let $Z_i = \text{multipercent}$. Then A2 becomes

$$E[Y_i | X_i, Z_i] = \delta_0 + \delta_1 X_i + \delta_2 Z_i. \text{ Recall that}$$

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{S_{xx}} = \frac{\sum_i (x_i - \bar{x})y_i}{S_{xx}}$$

$$\text{Thus } E[\hat{\beta}_1 | X's, Z's] = E\left[\frac{\sum_i (x_i - \bar{x})y_i}{S_{xx}} \mid X's, Z's\right]$$

$$= \sum_i \frac{(x_i - \bar{x})}{S_{xx}} E[Y_i | X's, Z's]$$

$$= \sum_i \frac{(x_i - \bar{x})}{S_{xx}} (\delta_0 + \delta_1 x_i + \delta_2 z_i)$$

$$= \sum_i \frac{\cancel{(x_i - \bar{x})}}{S_{xx}} \delta_0 + \sum_i \frac{\cancel{(x_i - \bar{x})} x_i}{S_{xx}} \delta_1 + \sum_i \delta_2 \frac{(x_i - \bar{x}) z_i}{S_{xx}}$$

$$= \delta_0 + \delta_1 \frac{S_{xz}}{S_{xx}} \text{ where } S_{xz} = \sum_i (x_i - \bar{x}) z_i \\ = \sum_i (x_i - \bar{x})(z_i - \bar{z})$$

where δ_1 is the true ceteris paribus effect of X_i on Y_i .

The bias is then $\delta_2 \frac{S_{xz}}{S_{xx}}$ and its sign depends on S_{xz} and δ_2 .

Note that $\beta_2 = \text{ceteris paribus effect of percentage of delivered homework on final score}$. We could argue that $\beta_2 > 0$. On the other hand s_{xz} can be computed from the data and is a sample measure of the correlation between attendance and homework delivery. Again, is not difficult to guess that this should be positive. Hence, the bias is positive and Atkinson's lack of inclusion of homework delivery percentage is biasing upwards the estimate of the effect of attendance.

(j) Any coherent explanation here is valid. For example, is clear that attendance doesn't really explain the final score, even if its effect is statistically significant. It is clear there are other factors at play that are more important to explain the final score. For example: hours studied, # of classes taken, GPA, etc. can all be used to predict final score.

Problem 2

(a) $\min_c \sum_{i=1}^n (X_i - c)^2$ is the LS problem.

The FOC is $\frac{dSS(c^*)}{dc} = -2 \sum_{i=1}^n (X_i - c^*) = 0$

$$\Leftrightarrow \sum_{i=1}^n X_i = \sum_{i=1}^n c^* = nc^* \Leftrightarrow c^* = \frac{1}{n} \sum_{i=1}^n X_i$$

Thus $c^* = \bar{X}$, i.e. the sample mean.

(b) We will show that $\bar{X}_n$ converges in L_2 and probability to a .

$$\text{Note: } E(\bar{X}_n) = \frac{1}{n} \sum_{i=1}^n E(X_i) = \frac{1}{n} \sum_{i=1}^n a = a = E(X)$$

$$\text{Using a computation from PS3, we have that } E[(\bar{X}_n - a)^2] = \text{Var}(\bar{X}_n) + (E(\bar{X}_n) - a)^2$$

$$\text{Note that } \text{Var}(\bar{X}_n) = \frac{\text{Var}(X)}{n} = \frac{\frac{2}{3} + a^2 - a^2}{n} = \frac{2}{3n}$$

$$\text{Since } \text{Var}(\bar{X}_n) = \frac{2}{3n} \xrightarrow{n \rightarrow \infty} 0 \text{ and } E(\bar{X}_n) = a \xrightarrow{n \rightarrow \infty} a$$

$$\text{Then } \lim_{n \rightarrow \infty} E[(\bar{X}_n - a)^2] = 0. \text{ Thus } \bar{X}_n \xrightarrow{L_2} a$$

and by Thm 4.1 $\bar{X}_n \xrightarrow{P} a$.

The convergence in probability could also be established by the WLLN. This results show that $\bar{X}_n$ is the appropriate large sample estimator of the parameter a .

(c) By the CLT; since $\text{Var}(\bar{X}_n) = \frac{2}{3n}$, we have.

$$Z = \frac{\bar{X}_n - a}{\sqrt{\frac{2}{3n}}} \sim N(0, 1)$$

$$(d) \Pr(|\bar{X}_n - a| \leq 0.05) = \Pr\left\{\left|\frac{\bar{X}_n - a}{\sqrt{\frac{2}{3n}}}\right| \leq \frac{0.05}{\sqrt{\frac{2}{3n}}}\right\}$$

$$= \Pr\left(|Z| \leq \frac{\sqrt{3n} (0.05)}{\sqrt{2}}\right)$$

$$= \Pr\left(-\frac{\sqrt{3n} (0.05)}{\sqrt{2}} \leq Z \leq \frac{\sqrt{3n} (0.05)}{\sqrt{2}}\right) \\ \approx \Phi\left(\frac{\sqrt{3n} (0.05)}{\sqrt{2}}\right) - \Phi\left(-\frac{\sqrt{3n} (0.05)}{\sqrt{2}}\right)$$

where Φ is the CDF of the $N(0, 1)$ distribution.

when $n = 3650$, we have that $\frac{\sqrt{3n} (0.05)}{\sqrt{2}} = 3.69966$

and the above probability is approximately:

$$\Phi(3.69966) - \Phi(-3.69966) \approx 0.9999 - 0.0001 \\ \approx 0.9998 \approx 1$$

(e) a is estimated with $\bar{X} = \frac{29,200}{3650} = 8$

call $\hat{a} = 8$. Then the pdf is estimated as

$$f(x; 8) = \frac{1}{4} [2 - 8 + x] I_{[8-2, 8]}(x) +$$

$$\frac{1}{4} [2 + 8 - x] I_{[8, 8+2]}(x)$$

$$= \frac{1}{4} (x - 6) I_{[6, 8]} + \frac{1}{4} (10 - x) I_{[8, 10]}$$

Then $\Pr(X > 21) = 1 - \Pr(X \leq 21)$

$$\Pr(X \leq 21) = \int_{-\infty}^{21} f(x; 8) dx$$

Since f is a pdf which integrates to 1 over the support $[6, 10]$, the above probability is equal to 1, since $(-\infty, 21]$ contains $[6, 10]$

Thus $\Pr(X > 21) = 1 - 1 = 0$.